The Emergence of Machine Labor Markets: An Adversarial Investigation into the Standardization and Financialization of AI-Completed Work

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Abstract

This monograph presents an adversarial investigation into whether work performed by autonomous AI agents will evolve into standardized, financialized economic markets comparable to commodities such as oil, electricity, freight, and cloud compute. The central thesis examines the conditions under which machine labor transitions from an input-priced resource (compute, tokens) to an output-priced commodity (verified units of completed work). We analyze twelve historical commodity markets and identify the institutional, technological, and economic preconditions for market formation: standardization of underlying units, credible benchmark pricing, trusted measurement infrastructure, sufficient heterogeneity and volatility to require hedging, and network effects that protect dominant institutions. We develop a rigorous framework for Standardized Machine Work Units (SMWUs) across eight task categories, and derive quality-adjusted (QAMLP) and risk-adjusted (RAMLP) pricing formulas that incorporate model API costs, compute, retries, human intervention, failures, compliance, and sovereignty constraints. We model ten market scenarios, evaluate a ten-stage evolution pathway, and construct a five-year business model with twelve revenue streams. An extensive falsification analysis tests fifteen objections. We conclude that machine labor markets are *conditionally inevitable*—likely only after specific thresholds in agent reliability, enterprise spending, standardization maturity, and capacity volatility are crossed. The most defensible initial position is as an independent benchmark administrator and data company (provisionally, *OnIndex*), with exchange infrastructure emerging through partnerships with regulated financial institutions rather than direct startup development. The decisive 90-day experiment tests whether token prices are weak predictors of final verified-outcome costs.

Keywords: machine labor, AI agents, commodity markets, benchmark pricing, standardization, financialization, sovereignty premium, outcome-based pricing, market formation

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1 Introduction

1.1 Research objectives

This monograph addresses a fundamental question in the economics of artificial intelligence: *will work performed by autonomous AI agents evolve into standardized and financialized markets comparable to oil, electricity, freight, cloud compute, advertising, and other critical resources?* We investigate this with an explicitly adversarial posture—actively attempting to disprove the thesis while rigorously identifying the conditions under which it might hold.

1.2 The central thesis

Compute is the input required to produce artificial intelligence; AI agents transform compute into productive economic work. If compute is developing commodity-market characteristics (benchmark prices, capacity markets, forward curves, reservations, hedging), the next economic transition may be from pricing *inputs* (GPU-hours, tokens) to pricing *outputs* (customer issues resolved, software tasks completed, contracts reviewed, insurance claims processed). The proposed entity, *OnIndex*, would become the independent measurement, benchmark-pricing, verification, and settlement layer for this machine labor, answering: *what is the current risk-adjusted market price of one verified unit of AI-completed work?*

1.3 Methodology

Our approach combines four lines of inquiry:

1. **Historical analysis.** Systematic study of twelve commodity markets to identify patterns in standardization, benchmark formation, and institutional development.
2. **Economic framework development.** Rigorous definitions of machine labor categories, Standardized Machine Work Units, and pricing methodologies.
3. **Scenario modeling.** Ten realistic market scenarios testing transaction structures and value creation.
4. **Falsification analysis.** Active attempts to disprove the thesis through fifteen structured objections, plus ten testable hypotheses with experimental designs.

2 Historical market analogies

Understanding whether machine labor will evolve into commodity markets requires systematic study of how other resources made this transition. For each of twelve markets we analyze: (i) the initial purchase unit, (ii) why markets were fragmented, (iii) how units were standardized, (iv) who created trusted benchmarks, (v) what caused spot markets to emerge, (vi) why forward contracts became useful, (vii) dominant institutions, (viii) network effects, (ix) which markets failed, and (x) relevance to machine labor.

2.1 Crude oil

In the early petroleum industry (1860s–1880s), oil was purchased in wooden barrels of varying sizes; quality was determined by visual inspection. Standard Oil introduced the 42-gallon barrel by the 1870s, though quality grading remained inconsistent. The American Petroleum Institute established standardized gravity measurements in the 1920s; chromatography in the 1950s enabled precise chemical analysis. *Standardization emerged only after the industry consolidated sufficiently to make universal standards economically rational for dominant players.*

West Texas Intermediate (WTI) emerged as the U.S. benchmark in the 1980s due to large stable production volumes, a storage hub at Cushing with pipeline connectivity, delivery-point specifications, and transparent pricing through NYMEX futures. *Benchmarks require physical infrastructure (storage, delivery points) as much as market design.* Spot markets developed in the 1970s after price deregulation and the collapse of the Seven Sisters' vertical integration. Futures (NYMEX WTI, launched 1983) became essential for producer/refiner hedging; open interest grew from negligible in 1983 to over 2 million contracts by 1990. S&P Global Platts became the dominant price-reporting agency; its Dated Brent assessment prices 70%+ of global crude because long-term contracts reference it, derivatives build on it, regulators adopt it, and new entrants face switching costs. *Benchmark network effects are stronger than exchange network effects because contracts reference benchmarks directly.* The Dubai and Bonny Light benchmarks declined when underlying physical volumes became insufficient or unreliable.

2.2 Refined fuels, electricity, natural gas

Gasoline and diesel developed standards (octane, sulfur, cetane) through associations and regulation; New York Harbor became the dominant U.S. gasoline delivery point. *Product standardization requires either regulatory mandate or industry-wide coordination.* Jet fuel standardized through ASTM aviation specifications, with Singapore emerging as the Asian benchmark.

Electricity began (1890s–1930s) with vertically integrated utilities and no wholesale market; the Public Utility Holding Company Act (1935) created regulated monopolies. Post-deregulation, markets fragmented by transmission constraints, regulatory differences, temporal variation, and non-storability. Standardization came through nodal (locational marginal) pricing, ISO/RTO formation (PJM, ERCOT, CAISO), standardized ISDA terms, and FERC Order 888 (1996). Multiple regional benchmarks emerged—no single global benchmark. *Electricity's non-storability and transmission constraints prevent global benchmark formation.*

U.S. natural gas transitioned from long-term take-or-pay contracts to spot markets after FERC Orders 436 (1985) and 636 (1992) mandated pipeline unbundling. Henry Hub became the national benchmark through pipeline interconnections, storage, and LNG coastal access; by 2000 over 70% of U.S. gas was priced relative to Henry Hub. LNG markets initially used oil-linked long-term contracts; spot LNG emerged after 2008 through U.S. shale exports and destination-clause removal. *Pipeline unbundling was necessary for spot formation—physical infrastructure separation enabled price discovery.*

2.3 Agriculture, freight, cloud compute

The Chicago Board of Trade (1848) developed standardized grain grades; the key innovation was not standardization itself but *transferable warehouse receipts*, enabling trading without physical movement. Soybean futures followed in 1936; options on futures in 1984. Soft commodities (coffee, cocoa, sugar) have international benchmarks but face quality variation and weather/political risk. *Standardization requires not just quality grades but transfer mechanisms.*

Shipping rates were historically bilateral. The Baltic Exchange (1744) developed the Baltic Dry Index (1985) and tanker indices through route definitions, vessel-size categories, and time-charter equivalents. *Even highly heterogeneous services can be standardized through route abstraction and category classification.* Forward Freight Agreements emerged in the 1990s as OTC derivatives cleared through LCH/SGX.

AWS Spot Instances (2009) used auction pricing for excess capacity; in 2017 AWS switched to provider-managed pricing, ending the auction. Compute standardized through VM instance types, vCPU/GB-RAM units, per-hour billing, and API-driven provisioning. Yet true secondary markets for capacity never emerged; no futures market exists. *Infrastructure control by providers may prevent full market development—providers prefer pricing power over market efficiency.*

2.4 Telecom, carbon, advertising, labor, FX, rates

Bandwidth exchanges (late 1990s: Enron Broadband, RateXchange) had largely failed by 2002. *Markets fail when quality varies by route, long-term contracts lock participants, carriers prefer private negotiations, and no settlement infrastructure exists.* Carbon markets formed through regulatory mandate (EU ETS 2005; California 2013), standardizing on the tonne-CO_2-equivalent unit with accredited verification and registries. EU carbon prices ranged from €3 (2013) to €100+ (2023), demonstrating policy-driven volatility.

Programmatic advertising developed real-time bidding with IAB ad-unit sizes, MRC viewability standards, and attribution models—yet opacity persists in viewability, bot traffic, and attribution; no independent benchmark exists for "one impression." BPO markets standardized on call-center seats per hour, claims per unit, data entry per record, and FTEs; gig platforms (Uber, Upwork) created spot labor markets through task decomposition, reputation, and settlement. FX achieved the highest standardization through international-trade hedging demand, standardized pairs, 24-hour trading, and electronic platforms (EBS, Reuters). LIBOR (1969–2023) became the floating-rate benchmark; manipulation scandals drove its replacement by SOFR/SONIA.

2.5 Comparative matrix

Table 1 summarizes the twelve markets. The cloud compute analogy is most relevant for understanding machine labor's market potential; BPO labor provides the best model for task-based pricing standardization.

Table 1: Historical commodity market comparison matrix.

Resource	Unit	Benchmark	Spot	Forward	Deriv.	Settlement
Crude oil	Barrel (42 gal)	WTI, Brent	Yes	Yes	Yes	NYMEX, ICE
Gasoline	Gallon	NYH RBOB	Yes	Yes	Yes	NYMEX
Electricity	MWh	PJM, ERCOT	Yes	Yes	Lim.	ISOs
Nat. gas	MMBtu	Henry Hub	Yes	Yes	Yes	NYMEX
Grain	Bushel	CBOT	Yes	Yes	Yes	CME
Freight	Voyage/time	BDI, BDTI	Index	Yes (FFA)	Yes	Baltic, LCH
Cloud	vCPU-hr	None	Yes	No	No	AWS/GCP/Azure
Carbon	tCO_2e	EU ETS, CCX	Yes	Yes	Lim.	ICE, EU
Ad imp.	Impression	None (CPM)	Yes (RTB)	No	No	Ad exchanges
BPO labor	Task/FTE	Per-task	Lim.	No	No	Outsourcers
FX	Curr. pair	EUR/USD	Yes	Yes	Yes	EBS, Reuters
Int. rates	Percent	SOFR, SONIA	Yes	Yes	Yes	CME, LCH

2.6 Key insights

Markets formed when: standardization was possible, sufficient scale existed, price volatility created hedging demand, independent verification was feasible, regulatory neutrality held, and network effects increased value with participation. **Markets failed when**: units were extremely heterogeneous, providers controlled infrastructure, relationships dominated, quality was unverifiable, or regulation prohibited cross-border/speculative trading. **Standardization precedes markets**; benchmarks require physical (or verification) infrastructure; forward markets emerge from volatility; dominant institutions achieve regulatory recognition; network effects protect incumbents; provider concentration can prevent full marketization.

3 Defining machine labor

3.1 The production function of AI work

Economically valuable AI work follows a multi-stage process:

$$\text{Compute+Data+Algorithms+Human Oversight} \longrightarrow \text{Intelligence} \longrightarrow \text{Agent Actions} \longrightarrow \text{Verified Outcome}$$

Each stage transforms the prior one into something more valuable but more specific. *There is currently no market pricing for verified outcomes; enterprise AI procurement remains at the input layer.*

3.2 Taxonomy

Intelligence and compute are separable: the same compute can produce different intelligence (different models on the same hardware), and the same intelligence can run on different compute (distillation, quantization, different providers). Model access can be bought per-token (API), per-hour (hosted inference), by license (private deployment), or self-hosted (weights + own infra). An AI agent is distinguished from software by autonomy, tool use, adaptation, and non-determinism—so agent pricing must account for probabilistic outcomes. *Capacity pricing (like cloud compute) differs fundamentally from outcome pricing (like BPO); the gap includes failures, retries, quality variation, and human intervention.*

3.3 Where outcome-based pricing holds and breaks

The claim that "companies want verified outcomes, not tokens" holds for customer support (resolutions are verifiable), claims processing, data entry, and document review. It breaks for creative work, research, exploratory coding, advisory services, and emergent tasks. *Outcome pricing is viable when success can be defined ex-ante, verification is cheaper than production, quality is measurable post-delivery, and retry is possible without catastrophic cost.* Most enterprise use will involve *hybrid* pricing: a subscription base for capacity, usage per task/outcome, quality-adjustment bonuses/penalties, and outcome guarantees with SLA refunds.

3.4 Is machine labor a coherent economic category?

Machine labor demonstrates coherence: Claude, GPT-4, and Llama are evaluated against each other for the same tasks; RFPs compare multiple providers; API prices have converged toward similar price-performance ratios; analyst firms track "AI agents" as a distinct category. Included are AI agents performing productive tasks, autonomous goal-directed systems, tool-using AI, and both digital and physical embodiments. Excluded are pure content generation, recommendation systems without agency, traditional RPA, and embedded AI without independence.

4 Standardized Machine Work Units

The central unsolved problem is standardization. Agent work varies enormously in task complexity, context, tool dependence, quality threshold, latency, error consequence, and domain specificity. Successful commodity standardization has relied on *grade bands* (not point specs), *reference methods*, *delivery-point specification*, and *third-party verification*—lessons directly applicable to machine labor.

4.1 Required specifications

A Standardized Machine Work Unit (SMWU) must specify fifteen parameters:

- Task category (support, software, legal, insurance, finance, cybersecurity, research, robotics)
- Difficulty level (1–5, with reference tasks)
- Expected output (format, completeness, dependencies)
- Success criteria (accuracy $\geq X\%$, coverage $\geq Y\%$)
- Quality threshold (Basic / Standard / Premium / Certified)
- Latency (real-time / interactive / batch / none)
- Max human intervention (autonomous / exception / QA / continuous)
- Acceptable retry rate (single / limited / unlimited)
- Security requirements (encryption, access, audit, SOC 2 / ISO 27001 / FedRAMP)
- Data residency (none / regional / country-specific / air-gapped)
- Model provenance (any / open-weight only / specific family / auditable training)
- Auditability (reasoning chain, decision traceability, tool records, intervention logs)
- Liability (provider / shared / buyer / insurance)
- Delivery period (immediate / standard / scheduled / batch)
- Verification process (automated / spot-check / full review / third-party audit)

4.2 Market evaluation

We evaluate eight candidate markets on repeatability, verifiability, volume, AI adoption, provider competition, price volatility, and standardization potential (Table 2). Customer support scores highest.

Recommendation. First market: customer support (highest standardization, volume, adoption, clear criteria). Second: financial operations (structured data, objective verification, lower regulatory complexity than insurance). Third: insurance claims (high volume, strong verification, manageable regulatory scope).

Table 2: Market standardization potential scoring (1–5 scale).

Market	Rep.	Ver.	Vol.	Adopt	Comp.	Volat.	Std.	Total
Customer support	5	5	5	5	4	3	5	**32**
Insurance claims	5	5	5	3	3	2	5	28
Financial ops	5	5	5	4	3	2	5	29
Legal review	3	3	4	3	3	3	4	23
Software eng.	3	5	5	3	3	4	4	27
Cybersecurity	3	3	4	3	4	3	3	23
Research	1	2	4	1	2	4	2	16
Robotics	2	4	3	1	1	4	3	18

5 Pricing and valuation frameworks

Concerns current AI pricing focuses on inputs rather than outputs, leaving buyers unable to predict the true cost of achieving outcomes. We develop the True Cost per Verified Outcome (TCVO), the Quality-Adjusted Machine Labor Price (QAMLP), and the Risk-Adjusted Machine Labor Price (RAMLP).

5.1 True Cost per Verified Outcome

The base formula aggregates every cost incurred to produce accepted work and divides by the number of accepted outcomes:

$$\text{TCVO} = \frac{C_{\text{model}} + C_{\text{compute}} + C_{\text{tools}} + C_{\text{infra}} + C_{\text{retries}} + C_{\text{human}} + C_{\text{failure}} + C_{\text{compliance}}}{N_{\text{accepted}}} \tag{1}$$

where:

$$C_{\text{model}} = \sum_{i=1}^{N_{\text{att}}} (T_{\text{in},i}\, P_{\text{in}} + T_{\text{out},i}\, P_{\text{out}}) \tag{2}$$

$$C_{\text{compute}} = \sum_{i=1}^{N_{\text{att}}} H_i\, R_{\text{gpu}} \tag{3}$$

$$C_{\text{tools}} = \sum_{i=1}^{N_{\text{att}}} \sum_{j=1}^{M_i} C_{\text{tool},i,j} \tag{4}$$

$$C_{\text{infra}} = C_{\text{orch}} + C_{\text{monit}} + C_{\text{log}} + C_{\text{store}} \tag{5}$$

$$C_{\text{retries}} = \sum_{r=1}^{N_{\text{ret}}} (C_{\text{model},r} + C_{\text{compute},r} + C_{\text{tools},r}) \tag{6}$$

$$C_{\text{human}} = H_{\text{rev}}\, R_{\text{rev}} + H_{\text{esc}}\, R_{\text{exp}} \tag{7}$$

$$C_{\text{failure}} = N_{\text{fail}}(C_{\text{att}} + C_{\text{rec}} + C_{\text{opp}}) \tag{8}$$

$$C_{\text{compliance}} = C_{\text{audit}} + C_{\text{cert}} + C_{\text{sec}} + C \tag{9}$$

C_{model} sums the per-attempt token cost (for N_{att} attempts); C_{compute} the GPU-hours; C_{tools} the external-API calls; C_{infra} orchestration/monitoring/logging/storage; C_{retries} equals or exceeds initial attempt costs for complex tasks; C_{human} the review and escalation labor; C_{failure} spans failed attempts, recovery/rollback, and the opportunity cost of delay; $C_{\text{compliance}}$ captures audits (SOC 2 \$50K–\$200K/yr; FedRAMP \$300K–\$1M+/yr), certification, security, and data-residency premiums (10–30% per region).

5.2 Quality-Adjusted Machine Labor Price

Not all outcomes are equal. Normalize to a standard quality:

$$\text{QAMLP} = \frac{\text{TCVO}}{Q_{\text{actual}}/Q_{\text{standard}}} \tag{10}$$

Example. Provider A: TCVO = \$2.00, accuracy 85%; Provider B: TCVO = \$2.50, accuracy 95%; standard 90%. Then $\text{QAMLP}_A = 2.00/(0.85/0.90) = \$2.12$ and $\text{QAMLP}_B = 2.50/(0.95/0.90) = \$2.37$—at standard quality, A is cheaper despite lower nominal accuracy. With multiple quality dimensions:

$$\text{QAMLP} = \text{TCVO} \prod_{k=1}^{K} \left(\frac{Q_{\text{std},k}}{Q_{\text{act},k}} \right)^{w_k}, \qquad \sum_k w_k = 1. \tag{11}$$

5.3 Risk-Adjusted Machine Labor Price

Different deployments carry provider-concentration, sovereign, latency, failure-mode, and model-version risks. Define risk factors $R_i \in [0, 1]$ with weight w_i:

$$R_{\text{conc}} = 1 - \frac{1}{N_{\text{providers}}} \tag{12}$$

$$R_{\text{sovereign}} = \begin{cases} 0 & \text{within jurisdiction} \\ 0.3 & \text{cross-border w/ residency controls} \\ 0.6 & \text{cross-border w/o controls} \\ 1.0 & \text{sanctioned jurisdiction} \end{cases} \tag{13}$$

$$R_{\text{latency}} = \sigma_{\text{latency}}/\mu_{\text{latency}} \tag{14}$$

$$R_{\text{failure}} = P_{\text{catastrophic}} \cdot I_{\text{catastrophic}} \tag{15}$$

$$R_{\text{version}} = 1 - \text{stability}_{\text{model}} \tag{16}$$

The risk-adjusted price is

$$\text{RAMLP} = \text{TCVO} \left(1 + \sum_{i=1}^{n} w_i R_i \right), \qquad \sum w_i = 1. \tag{17}$$

If failure insurance is available at premium P_{ins} with expected payout $\mathbb{E}[\text{payout}]$:

$$\text{RAMLP}_{\text{insured}} = \text{TCVO} + P_{\text{ins}} - \mathbb{E}[\text{payout}]. \tag{18}$$

5.4 Benchmark methodology

We recommend a *trimmed volume-weighted average*: exclude the top and bottom 10% by price (outlier/wash-trade control), volume-weight the remainder, apply quality adjustment pre-aggregation, and publish on a fixed cadence. Manipulation resistance requires transaction verification (task hashes on immutable ledgers, independent cross-checks), multi-source aggregation (buyer, provider, platform, third-party), outlier statistical filters, transparency (public methodology, external audit rights, anonymous-but-disclosed sources), and governance (an independent oversight committee with market-participant and regulatory representation).

6 Jurisdiction and the sovereignty premium

AI systems are subject to developer jurisdiction, training-data law, cloud-provider control, and buyer compliance. Three ecosystems dominate—U.S. (OpenAI, Anthropic, Google, Meta, xAI), China (Baidu, Alibaba, Tencent, 01.AI, Moonshot), Europe (Mistral, Aleph Alpha, DeepMind)—under different regulatory regimes. Open-weight models (Llama, Qwen, Mixtral, Yi) complicate the landscape: they enable cross-jurisdiction deployment, but origin still matters for compliance and trust.

6.1 The sovereignty premium

The *sovereignty premium* is the additional price paid to ensure AI work satisfies national, security, data, or model-origin requirements. Sources include data residency, model-origin preference, security certification (FedRAMP/EUCS), air-gapped deployment, audit/oversight rights, and supply-chain verification. Estimated premiums (% of base price):

Table 3: Estimated sovereignty premiums.

Requirement	Min	Typical	Max
Data residency (EU)	10%	20%	40%
Data residency (multi-region)	20%	35%	60%
FedRAMP moderate	30%	50%	100%
FedRAMP high	50%	100%	200%
Air-gapped deployment	100%	200%	400%
Domestic model only	20%	50%	100%

Basis: cloud-region premiums 10–30%, FedRAMP compliance overhead 50–100%, air-gapped infrastructure 2–4×.

6.2 Index grades and supply-chain analysis

OnIndex should publish eight differentiated indexes: Global Best-Price; U.S. Sovereign; China Domestic; European Data-Resident; Open-Weight; Private-Cloud; Air-Gapped; Defense-Eligible. *The complete execution and data supply chain matters more than model nationality*: a Chinese open-weight model on U.S. infrastructure with U.S. data residency may be more acceptable to U.S. buyers than a U.S. closed model on Chinese infrastructure—mirroring the TikTok-vs-AWS-China debate. Buyers of this data include multinational enterprises, government procurement, regulated industries, AI providers, and investors. Governments would use such indexes to compare national AI competitiveness, assess foreign dependence, inform industrial policy, and negotiate trade agreements.

7 Market scenarios

We model ten realistic transactions. Selected summaries:

Scenario 1—Retailer reserving AI support capacity. National retailer, 500M revenue, 2M interactions/yr; 500,000 resolutions during Black Friday week at \$1.20/resolution vs. \$8.50 for humans (90% accuracy, 95% CSAT, $< 2\%$ escalation). OnIndex benchmark \$1.15; weekly invoicing; verification by post-interaction sampling. Without OnIndex neither side can confirm fair pricing or competitiveness.

Scenario 2—Bank comparing models for compliance. 50,000 documents/yr across Claude, Qwen, Mistral, internal (95% recall, 90% precision). Sovereignty-adjusted prices: U.S. $4.50/doc; EU data-resident $5.20; open-weight self-hosted $3.80; China-hosted ineligible.

Scenario 3—Software budgeting human vs. machine labor. Human engineer $158/task (95% success) vs. AI agent $77/task (70% success, after token cost, $30 review, $15 rework). TCVO reveals the true trade-off.

Scenario 4—Government measuring sovereign AI costs. Standardized task basket: Global $1.00; U.S. Sovereign $1.35 (+35%); EU Data-Resident $1.45 (+45%); Defense-Eligible $2.80 (+180%).

Scenarios 5–10. Startup certification (independent TCVO testing); enterprise dynamic provider routing via price/quality data with 15–30% savings; insurer underwriting agent-failure insurance using OnIndex actuarial data (2.3% failure rate, $1,150 premium per $50K coverage); seasonal capacity reservation (forward curve, spot $0.80 → forward $1.00, 25% premium); bank financing a provider at SOFR+400bp vs. SOFR+800bp without benchmark; hospitality comparing human ($8.00/delivery) vs. robotic ($7.50) labor.

7.1 Cross-scenario analysis

All scenarios benefit from independent price discovery, quality verification, risk quantification, and contract standardization. Immediate use cases (benchmarking, certification) sit alongside advanced applications (insurance, financing, forwards); all exhibit significant transaction friction without independent indexing.

8 Market evolution

We propose a ten-stage pathway (Table 4) with declining probability at advanced stages.

Table 4: Market evolution stage summary.

Stage	Timeline	Prob.	Rev. model	Key risk
Measurement	2024–26	90%	SaaS	Data sharing
Benchmarking	2026–28	80%	Subscriptions	Standardization
Certification	2027–29	70%	Testing fees	Gaming
Index licensing	2028–30	60%	Licensing	Legal
Spot procurement	2029–31	50%	Transaction fees	Liquidity
Capacity reserve	2030–32	50%	Reservation fees	Scarcity
Forward contracts	2032–35	40%	Arrangement	Volatility
Insurance/financing	2031–34	50%	Data licensing	Moral hazard
Exchange	2035–40	25%	Trading fees	Regulation
Derivatives	2040+	15%	Index licensing	Commoditization

Stages 1–4 (measurement through index licensing) are a credible business trajectory; *stages 5–8* (spot through insurance) require conditions that may not materialize; *stages 9–10* (exchange, derivatives) are speculative. The prudent strategy focuses on 1–4 while maintaining optionality for later stages through partnerships with regulated financial institutions.

9 Business model

Twelve revenue streams span immediate, later, and speculative credibility. **Immediately credible**: enterprise market-data subscriptions ($20K–$100K/yr); procurement benchmarking ($50K–$300K/engagement); provider testing and certification ($50K–$200K/cert). **Later-staged viable**: index/API licensing ($10K–$50K/yr); government/sovereign intelligence ($500K–$2M/yr); procurement transaction fees (3–5%); verification/settlement fees (1–2%); capacity-reservation fees (5–10%). **Speculative**: insurance-data licensing ($100K–$500K/yr); benchmark licensing to exchanges ($1M–$10M/yr); financing/underwriting data; robotic-labor data.

9.1 Five-year projection

Table 5: Five-year business model projection (ARR in USD).

Stream	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5
Enterprise subscriptions	0.3M	1.2M	2.5M	4.0M	6.0M
Certification testing	0.2M	0.8M	1.5M	2.5M	4.0M
Procurement consulting	—	0.5M	1.0M	1.5M	2.0M
Index licensing	—	—	0.5M	1.5M	3.0M
Transaction fees	—	—	—	1.0M	3.0M
Government contracts	—	—	—	—	1.0M
Total ARR	**0.5M**	**2.5M**	**5.5M**	**10.5M**	**19.0M**
# Enterprises	10–15	40–60	100–150	200–300	400–600
# Providers	4–6	15–25	40–60	80–120	150–250
Headcount	8–10	20–25	40–50	70–90	120–150

Unit economics target LTV/CAC $\geq$ 3:1. Margins: subscriptions 80–90%; certification 60–70%; consulting 40–50%; transaction fees 70–80%. Path to $1M ARR: 15–20 enterprises or 10–15 + 5–8 certifications within 12–18 months. Conditions for a $1B outcome (10–20% probability): machine labor becomes a significant Fortune 500 cost category, OnIndex achieves dominant benchmark position, marketplace reaches $1B+ GMV, index licensing is widely adopted, and expansion to 3–5 international markets.

10 Competitive landscape

Competitors cluster into: model benchmarking (Artificial Analysis, LMSYS, MLCommons), AI observability (LangSmith, W&B, Arize), model routing (Martian, OpenRouter, Together AI), AI gateways (Kong, Cloudflare), GPU exchanges (CoreWeave, Lambda Labs), and agent marketplaces (Relevance AI, LangChain). OnIndex is differentiated by its focus on *verified outcomes* rather than model capabilities, its independent third-party position, standardized SMWU definitions, and the true-cost methodology.

11 Regulation and market design

Benchmark manipulation has occurred in LIBOR, ISDAfix, and WM/Reuters FX. Machine labor benchmarks face similar risks where expert judgment enters price formation, participants are concentrated, and financial incentives are large. The EU Benchmarks Regulation (BMR) requires authorization, methodology transparency, governance, and conflicts-of-interest management. *The most defensible structure is an independent data and certification company, licensing benchmarks to regulated exchanges rather than directly operating exchange infrastructure.*

12 Falsification analysis

We test fifteen objections (Table 6). The two critical threats to the entire thesis are Objection 3 (compute remains the only relevant commodity) and Objection 7 (buyers will not share data); both require empirical validation rather than further argument.

Table 6: Falsification objection summary.

Objection	Threat	Mitigation
1 Heterogeneity	Med. (Exchange)	Grading system
2 Price collapse	High (Forwards)	Focus on core
3 Compute only	**Critical** (Entire thesis)	Survey validation
4 Long-term contracts	Med. (Marketplace)	Index licensing
5 No scarcity	High (Forwards)	Accept limitation
6 Vertical integration	Med. (Share)	Independence
7 No data sharing	**Critical** (Entire thesis)	Pilot required
8 Quality unverifiable	Med. (Settlement)	Narrow scope
9 Human necessary	Low (Autonomy)	Include in cost
10 No liquidity	High (Forwards)	Accept limitation
11 Remains software	Med. (Commodity)	Focus on layers
12 Incumbents win	High (Share/Margins)	Speed/focus
13 Government blocks	Med. (Global)	Regional strategy
14 Outcome pricing removes index	Low (Marketplace)	Limited threat
15 Premature	High (Exchange)	Accept limitation

Verdict. Objections 1, 4, 6, 8, 9, 11, 13, 14 are manageable for the core business; 2, 5, 10, 12, 15 are high threats to advanced stages; 3 and 7 are critical and gate the entire thesis. The prudent conclusion: stages 1–4 are viable with proper execution; stages 5–10 face substantial obstacles that may prove insurmountable; the exchange vision appears premature and possibly incorrect.

13 Empirical research design

We propose ten testable hypotheses; the keystone is:

H1. *Token prices explain* $< 60\%$ *of variance in True Cost per Verified Outcome.*

The decisive experiment:

1. Integrate with 3 design-partner enterprises (weeks 1–2).
2. Collect data on 5,000+ tickets per partner (weeks 3–8).
3. Calculate TCVO per ticket; analyze token–TCVO correlation (weeks 9–10).
4. Present findings and measure willingness-to-pay (weeks 11–12).

Success criteria. Token–TCVO correlation < 0.6; $2\times+$ TCVO variation across providers for same task types; 2+ design partners commit to paid subscription; a clear path to 10 paying customers within 6 months.

Evidence to continue. 90-day experiment meets criteria; 2+ paying customers within 6 months; clear path to $1M ARR within 18 months.

Evidence to abandon or pivot. Token–TCVO correlation > 0.8 (token prices are sufficient); enterprises refuse to share data even under NDA; no paying customers after 12 months; cloud providers launch credible competing services. If data sharing proves impossible but cost-optimization expertise demand survives, pivot to consulting/advisory.

Supporting hypotheses: H2 model rankings change with full cost accounting; H3–H4 prices vary across jurisdictions and sovereignty premiums persist; H5 at least one task category can be standardized (inter-rater reliability $\kappa > 0.7$ among 10 experts classifying 1,000 tickets); H6 enterprises will pay for neutral comparisons; H7 demand exhibits seasonality; H8 benchmark ownership produces defensibility; H9 the dataset is the most valuable asset; H10 the exchange is built through partnerships.

14 Limitations

- **Data constraints.** Limited empirical data on actual enterprise AI costs due to confidentiality.
- **Rapid change.** The AI market evolves faster than the research cycle; conclusions may become outdated.
- **Geographic bias.** Research weighted toward U.S. and European markets.
- **Survivorship bias.** Historical analysis focuses on successful markets; failed markets are understudied.
- **Provider coordination.** Uncertainty about how model providers will respond to commoditization pressure.

15 Conclusion

15.1 Assessment of four conclusions

We evaluate four possible conclusions:

- **A. Structurally inevitable** (highly likely under current trends)—*not supported*; falsification reveals multiple critical obstacles.
- **B. Conditionally inevitable** (likely only after specific thresholds)—*supported*; stages 1–4 are viable contingent on enterprise data sharing, 2×+ TCVO variation, SMWU adoption, and regulatory neutrality.
- **C. Plausible but premature** (useful index, exchange too early)—*partially supported*; stages 1–3 are near-term viable, stage 4 needs 2–3 years, stages 5–8 are premature.
- **D. Incorrect market analogy** (remains software/services)—*partially supported for advanced stages*; stages 9–10 may be an incorrect analogy, but stages 1–3 are valid regardless of full commoditization.

The evidence supports **Conclusion B** for stages 1–4, with elements of **C** for stages 5–8 and **D** for stages 9–10.

15.2 Final statement

Machine labor markets are likely to develop through measurement, benchmarking, and certification because enterprises demonstrably struggle with AI cost visibility, providers seek credible differentiation, precedent exists in cloud cost optimization and security certification, and the value proposition is clear. Forward markets and exchange infrastructure face substantial obstacles: price volatility may be insufficient, provider concentration may prevent formation, liquidity may never reach institutional thresholds, and regulatory burden may exceed value. *The weakest assumption is that enterprises will share detailed outcome data with an independent third party*—that requires validation through the 90-day pilot, not further analysis.

The strongest version of OnIndex answers one expensive enterprise question: *what is the true cost per verified outcome, accounting for all hidden costs and quality variations?* The time for research has passed; the time for experimentation has begun.

A Standardized Machine Work Unit specification template

Table 7: SMWU specification parameters.

Parameter	Specification
Task category	[Category]
Task difficulty level	[1–5]: [Reference Task ID]
Expected output	[Format and requirements]
Success criteria	Accuracy $\geq X\%$, Coverage $\geq Y\%$
Quality threshold	[Basic/Standard/Premium/Certified]
Latency requirement	[Real-time/Interactive/Batch/None]
Max human intervention	[Autonomous/Exception/QA/Continuous]
Acceptable retry rate	[Single/Limited/Unlimited]
Security requirements	[Compliance framework]
Data residency	[Constraint specification]
Model provenance	[Open/Closed/Specific]
Auditability	[Logging requirements]
Liability	[Responsibility assignment]
Delivery period	[Timing specification]
Verification process	[Validation method]

B Model pricing compilation

As of mid-2026:

Table 8: Model API pricing.

Model	Input (per 1K tokens)	Output (per 1K tokens)
GPT-4o	\$0.005	\$0.015
Claude 3.5 Sonnet	\$0.003	\$0.015
Llama 3.1 405B (hosted)	\$0.002	\$0.002

GPU pricing: AWS p4d.24xlarge (8×A100) \$32.77/hr; GCP a2-ultragpu-8g (8×A100) \$40.80/hr; Lambda Cloud 1×A100 \$1.99/hr.